STEAMROOT · PRODUCT REQUIREMENTS AND SPECIFICATIONS

SteamRoot product requirements and specifications

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DOCUMENT CONTROL

Document ID	STR-PRS-001	Version	0.8
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Migrated from STR-REQ-001 to STR-PRS-001 (kit 1.9.1 upgrade): R1 to R12 become PR-001 to PR-015 with the same targets, each traced to STR-URS-001; status from STR-CAL-001 v0.5 (gas-pass baffles): PR-004 at risk to not met, PR-005 met to at risk, PR-008 met to at risk (battery); PR-016 to PR-018 proposed, awaiting Amish; after the independent review: PR-011 at risk to not met, PR-005 66.1 %, PR-010 at 3.2 m, PR-009 basis open
0.8	2026-10-02	Amish Chadha	Status from STR-CAL-001 v0.4 (evaporator bank, dampers, seal pot overflow, ballast handles): R4 not met to at risk, R5 not met to met, R11 mass met to at risk; R4 restatement to 4.5 kg/m2 proposed, awaiting Amish
0.6	2026-10-02	Amish Chadha	R4 to be restated after the evaporator bank study and R5 study to keep R7

			(decided 2026-10-02); authority named; no status changed
0.5	2026-10-01	Amish Chadha	TRL 3 status updated for the constructable design (STR-DDR-003, STR-CAL-001 v0.3); R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (two hoods in R2, $1,800 budget in R12); correct the wood heating value in R4; add TRL 3 status from STR-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2, including pressure and safety requirements
0.1	2026-09-24	Amish Chadha	Initial scaffold

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What the design must do, each requirement traced to a user need in STR-URS-001. The targets are those of STR-REQ-001 v0.7 as decided by Amish on 2026-09-25 and 2026-10-02, carried over unchanged; R2, R6 and R11, which each held two targets, are split into two rows and the old number is in "Was". Status is from STR-CAL-001 v0.5 (`docs/04-calcs/01-sizing.md`), which carries the gas-pass baffles fitted on 2026-10-24 under Amish's standing instruction of 2026-10-23 (STR-DEC-001 S1), and the 3.2 m chimney outlet and latching pump trip of 2026-10-25 (S3, S4). Targets remain proposals to review with the co-design partners. Of the 18 rows, none is withdrawn; 2 are not met (PR-004, with a restatement proposed, and PR-011, with a basis proposed), 1 is not met for one operator and proposed to Amish (PR-016), 5 are at risk, 1 is over the value-engineering target, 1 is to verify at TRL 4 and 8 are met on paper or by design. Safety requirements (PR-006, PR-007, PR-009, PR-010, PR-013, PR-017) take priority over performance requirements if they conflict. Method: STANDARDS section 22.

> **Safety:** SteamRoot is a fired steam generator. Meeting these requirements on paper does not make it safe or legal to build. A written ruling from the local boiler authority is an open prerequisite for any detail design or build.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-001	UR-001	Performance	Pasteurize soil to working depth	Soil shall reach 70 °C (158 °F) or more for 20 to 30 min at 10 to 15 cm depth across at least 80 % of the hood footprint	70 for 20 to 30 min, 80 % of footprint	min	°C, min, %	Analysis (CAL-001 section 2), TRL 3; thermocouple probes at 15 cm, TRL 4	To verify at TRL 4 (1D model holds 70 °C at 15 cm if the front reaches 16.5 cm; STEM S11 fails: in soils finer than sand the steam cannot pass at full flow under the hood; bed preparation and hold-down proposed, DEC #18)
PR-002	UR-002	Performance	Treatment rate at 15 cm	Two hoods used alternately shall treat at least 1.85 m2/h (about 1.9) at 15 cm depth (restated by Amish 2026-09-25, STR-DDR-002)	1.85	min	m2/h	Analysis (CAL-001 section 3), TRL 3; timed trial, TRL 4	At risk: 1.88 m2/h; about 1.5 to 1.6 m2/h net of edge strips, overlap and hose losses (STEM M2)
PR-003	UR-003	Performance	Steam output	The generator shall make at least 30 kg/h of saturated steam, continuously, from one firebox	30	min	kg/h	Analysis (CAL-001 section 4), TRL 3; feed metering, TRL 4	Met on paper: 30 kg/h at 32.2 kW firing
PR-004	UR-004	Performance	Wood per m2	Wood use shall be at most 4 kg at 20 % moisture (about 14.5 MJ/kg) per m2 at 15 cm (target kept 2026-09-25; to be restated,	4.0	max	kg/m2	Analysis (CAL-001 section 4), TRL 3; fuel weighing, TRL 4	Not met: 4.36 kg/m2 with the baffles and dampers set (4.44 with the dampers wide open); restatement to 4.5 kg/m2 proposed,

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				decided 2026-10-02)					awaiting Amish (DEC #12)
PR-005	UR-004	Performance	Fuel-to-steam efficiency	At least 65 % of the fuel energy shall reach the water as steam, with the economizer and evaporator bank	65	min	%	Analysis (CAL-001 section 4), TRL 3; flue temperature, oxygen and fuel weighing, TRL 4	At risk: 66.1 % at λ = 1.5 with the gas-pass baffles, a margin smaller than its uncertainties (62.5 to 66.9 % i CAL Table 6a; 59.8 % without baffles)
PR-006	UR-005	Safety	Header pressure and open vent	Normal working pressure at the steam header shall be at most 0.1 bar gauge (1.5 psi), with an open vent to atmosphere that cannot be closed or isolated (redefined by Amish 2026-09-25)	0.10	max	bar gauge	Analysis (CAL-001 section 5); design review, TRL 3	Met on paper: 0.033 bar; seal limit 0.084 bar; vent cannot be isolated
PR-007	UR-005	Safety	Stored water	The heated section shall hold at most 8 L of water, with no closed steam drum	8.0	max	L	Analysis (CAL-001 section 6), TRL 3	Met on paper: 7.3 L flooded
PR-008	UR-006	Usability	Run between refills	One feed water fill, and one battery charge, shall last at least 3 h of steaming	3.0	min	h	Analysis (CAL-001 section 7), TRL 3; timed run, TRL 4	At risk: water 4.2 h; the 20 Ah battery runs to about 100 % depth of discharge per fill at an assumed 4 A pump (STEM E3; DEC #16)

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-009	UR-005	Safety	Certified relief valve	A certified steam relief valve set at 15 psi (1.03 bar) or less shall pass both the full steam output and the dry-coil refeed flash, as a second line of protection behind PR-006 (reworded by Amish 2026-09-25)	15 psi set; capacity at least the refeed flash	max set, min capacity	psi, kg/h	Valve rating and capacity check (CAL-001 section 6), TRL 3	Met on paper: rated 170 kg/h against a 141 kg/h flash on a 30 s basis; the flash basis is open (STEM S16, DEC #20); confirm on the valve bought
PR-010	UR-007, UR-013	Safety	Operator and site protection	The chimney outlet shall be at least 2.2 m above ground with a spark arrestor of 6 mm mesh or finer; the fire shall be outdoors only; steam shall be diverted to the vent before a hood is lifted; hood handles shall stay at 60 °C or cooler	2.2 m; 6 mm; 60 °C	min; max; max	m, mm, °C	Design review, TRL 3; surface temperature check, TRL 4	Met by design: 3.2 m (raised for draft, DEC S3), 6 mm mesh, hood skin 23 °C
PR-011	UR-008	Packaging and transport	Towed mass	The machine as towed, with the feed tank and seal pot drained, shall weigh at most 500 kg (restated	500	max	kg	Analysis (CAL-001 section 7 and the model), TRL 3; weighbridge, TRL 4	Not met: 509 kg with both hoods aboard (643 kg full; 447 kg without the hoods, which ride on a second vehicle for the prototype; basis

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				by Amish 2026-09-25)					proposed in DEC #13)
PR-012	UR-009	Packaging and transport	Parts cost	Parts cost shall be at or under the USD 2,200 value-engineering target, a hypothetical control target, not a limit (Amish 2026-10-01)	2,200	max	USD	Priced BOM, TRL 3	Over target: USD 2,720 (USD 520 over; the overrun to USD 2,600 was accepted by Amish on 2026-10-03, the USD 120 of STEM fixes is new)
PR-013	UR-005	Safety	Coil inlet pressure	Normal working pressure at the coil inlet (now the evaporator bank inlet) shall be at most 0.15 bar gauge (2.2 psi)	0.15	max	bar gauge	Analysis (CAL-001 section 5), TRL 3	Met on paper: 0.129 bar
PR-014	UR-002	Performance	Treatment rate at 5 cm	Two hoods used alternately shall treat at least 3.4 m2/h at 5 cm depth (weed seed bank only)	3.4	min	m2/h	Analysis (CAL-001 section 3), TRL 3	At risk: 3.41 m2/h
PR-015	UR-008	Packaging and transport	Overall width	Overall width as towed shall be at most 1.5 m	1.5	max	m	Model and drawing STR-DWG-002, TRL 3	At risk: 1.48 m
PR-016	UR-010	Usability	Hood handling	No person shall lift more than 23 kg when a hood is moved between settings (revised	23	max	kg per person	Analysis (STEM B4), TRL 3; handling trial, TRL 4	Not met for one person: 31.3 kg hood, handles 1.12 m apart; a two-person share of 15.6 kg is borderline once the reach is counted. Wheel

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				NIOSH lifting equation load constant), ballast weights lifted off first					kit recommended; proposed, awaiting Amish (DEC S2)
PR-017	UR-011	Safety	Two independent means against pressure	The steam side shall be designed for operation only at atmospheric pressure and carry two independent means to prevent the build-up of pressure (Texas Health and Safety Code 755.022(a)(4)), subject to the authority's written ruling	2	min	independent means	Design review, TRL 3; written ruling from TDLR before any build	Met by design (water seal and certified relief valve); ruling not yet obtained. Proposed, awaiting Amish (DEC #17)
PR-018	UR-012, UR-014	Physical architecture	Workshop build	The machine shall be built from stock steel sections, stock stainless tube and catalogue safety parts with the tools of a farm workshop and a competent welder, following STR-BLD-001	Stock and catalogue parts only	none	none	Inspection of the model, BOM and build plan, TRL 3	Met by design (STR-BLD-001; `cad/fidelity.json` D3). Proposed, awaiting Amish (DEC #17)

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- Soil to 15 cm depth: bulk density 1,300 kg/m³, gravimetric water content 20 % (dry basis), starting temperature 15 °C. The soil is modelled with a condensation front: soil above the front reaches 100 °C, and the front must reach 16.5 cm so that 15 cm stays at 70 °C or more through a 25-minute hold (STR-CAL-001, section 2).
- Specific heat of dry mineral soil 0.84 kJ/(kg K), of water 4.18 kJ/(kg K).
- 70 °C for 20 to 30 minutes sits inside the extension guidance for weed seeds and soil-borne pathogens (STR-STM-001, B1). Some hard-seeded weeds and heat-tolerant viruses need higher temperatures; this is accepted.
- Wood at 20 % moisture (wet basis) has a lower heating value of about 14.5 MJ/kg for a typical hardwood analysis.
- Pressure thresholds for boiler and pressure vessel rules vary by jurisdiction. PR-006, PR-009 and PR-017 are set to stay within the Texas exemption for boilers designed for operation only at atmospheric pressure, but compliance must be confirmed: the written ruling from the Texas Department of Licensing and Regulation's boiler program is an open prerequisite.
- PR-011 assumes the machine is towed with the feed tank and seal pot drained and filled from a water supply at the site.
- The PR-009 capacity check uses the maker's published capacity for the Watts Series 315, 3/4 in, 15 psi valve (375 lb/h); it must be confirmed for the valve actually bought.
- PR-002 and PR-014 assume two hoods used alternately, a 25-minute hold, 2 minutes to reset a hood and 1 minute of vented steam each time the hose is moved.
- PR-008 assumes a pump current of 4 A until a pump is chosen; the datasheet value replaces it.